Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Towe, Bruce Ph.D.		
Project Title:	A Microfluidic Bioreporter for Exploratory Probes		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bruce.towe@asu.edu	**Fax:**	FY 602 727-7624
PI Organization Type:	UNIVERSITY	**Phone:**	602 965-4116
Organization Name:	Arizona State University		
PI Address 1:	Room ECG 324, College of Engineering and Applied Sciences		
PI Address 2:	Box 879709		
PI Web Page:			
City:	Tempe	**State:**	AZ
Zip Code:	85287-9709	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Towe, Bruce		
COI Name (Institution):	Pellis, Neil (NASA - JSC)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	An innovative approach to a compact bioreporter chip is being investigated whose function will be to act as a probe for spacecraft, extraplanetary, and earth environments. It will give information that is difficult or impossible to derive from combinations of physical measurements since it gives the added relevance of environmental effects on life. The proposed hardware device will support a small (50 microliter) culture of bacteria in a compact plastic package that would fit into a shirt pocket. Microfluidics controlled by a microprocessor will feed and maintain the bacteria and monitor changes in their metabolic biochemical profiles as an indicator of environmental stressors. The device employs an integrated microchemistry system to monitor culture biochemistry and control feeding based on this. By comparing cellular metabolic profiles in terms of rates and quantity of uptake of nutrients and its production of primary and possibly secondary metabolites, this work will profile cellular responses to environmental stressors. Living bacteria as bioreporters can present a wide variety of sensitivities and unique responses to environmental stresses. Applications might include monitoring radiations, magnetic and geomagnetic fields, response to microgravity, as well as to gaseous,		

Task Description:	aqueous, or liquid-carried filtrates of soil or other aspects of the environment. The objective of this project is to: 1. Employ microfluidic technologies to create perfused cell microcultures on a compact monolithic plastic substrate. 2. Integrate the microculture with a series of microfluidic-based biochemistry sensors to monitor the cellular uptake of nutrients and products of cellular metabolism. 3. Implement intelligent computer control and data capture from the bioreporter. 4. Characterize the biomolecular signatures of the test cell lines by exposure to control as well as challenging environments. This is an interdisciplinary project involving bioengineering and cell science. We are interested in the development and use of bioreporters as both a probe and a tool that would ultimately allow the posing of multiple different scientific hypotheses of importance not only in space environments but earth-based applications as well. Biochemistry monitoring of bacterial metabolism or possibly other types of cells is a key aspect of the envisioned bioreporter system. Microchemistry sensors employed in this work are based on a color-changing or fluorescence-producing chemical reaction that occurs when a dialyzed sample of cell culture effluent is mixed with a specific reagent. The microfluidic hardware configuration of sampling, mixing, and the measurement of color, absorbance, or fluorescence can be adapted to measure the concentration of nearly any low molecular weight biochemical for which there exists a sensitive optical test. Besides glucose, oxygen, carbon dioxide, and pH, the system has the possible ability to measure glutamine, other sugars, urea, uric acid, lactic acid, acetic acid, ethanol, ammonia and other biochemicals. Man's venture into new and little understood environments of space and other planets presents a host of unknowns about how life will react and adapt. Simple physical instruments cannot easily determine the effect of environments on the living organisms because of life's complexity, ability to self-repair, and to change in order to accommodate to the environment. Instrumented probes will be used in space and extraplanetary venues to measure the physical aspects of the environment such as its gravity, atmosphere, radiation environment, and ultimately the suitability of its soil to support life. However we have no good way of predicting the aggregate of effects of multiple conditions of which none individually is lethal. To assess these environments, bioreporter systems will be a key element in the human exploration and development of space. A bioreporter combines living cells such as bacteria or mammalian cells with one or more sensors so as to provide information on how the environment affects cellular processes. Bioreporters can give insight into how the environment's chemical composition, atmosphere, light, and radiation fields interact with the complexity of life processes rather than just a single environing physical condition. When used to monitor and then alarm upon a specific circumstance or hazardous condition, a bioreporter would be called a biosentinel. Both bioreporters and biosentinels are enabling technologies that have the potential to serve an essential role in the initial and intermediate phases of the exploration for planetary missions. Their advantages lie in their direct relevance to human exploration of space, compact size, cost effectiveness, and adaptability to meet specific exploration objectives and goals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Cell-culture on-a-chip systems have been constructed that contain all of the necessary support systems for feeding, monitoring, and extracting waste products for 35 ul cultures of E.Coli bacteria. These devices are self contained and self-monitoring and use a unique microdialysis system implemented in microfluidics to extract cell waste products. These devices are made from acryllic by computer controlled micromachining and so can be made reproducibly and in large numbers to implement arrays of cultures. They have on-board micropumps, microvalves, sensors, and control circuitry and connect to laptop computers to report growth plots and trends. These devices have supported cyclic bacterial growth over week-order durations. Each microculture incorportates optical sensors that monitor cell culture optical density and pH and so control the delivery of media and extract waste. Some versions perform CO2 monitoring. The experimental goals have been upgraded to incorporate the use of lux-fused E Coli bacteria having reporter genes that will be inducible by environmental stressors. To use these reporters we have integrated two optical sensors in to the microculture system design. These include a fluorescence microsensor for green fluorescent protein (GFP) and a high performance optical sensor for luminescence. We have shown that natively bioluminescent organisms can readily be monitored optically for growth over week periods. This recent work expands the potential applications of the microculture system over what we originally planned and envisioned. We can now continuously monitor not only metabolic changes in growing cultures but genomic changes. More powerfully, engineered bacteria can couple light production directly to specific stress response pathways on the molecular level. By monitoring activation of known bacterial stress responses this bioporter approach is designed to use living cells as a way of characterizing the specific nature of hostile environments. The ultimate aim is to create very compact arrays of microcultures wherein each contains multiple bacteria that are variously responsive to a specific stress, to construct a ‘fingerprint’ of tested environments.
Bibliography Type:	Description: (Last Updated: 06/19/2005)
Articles in Peer-reviewed Journals	Cooney C. and Towe, B. "A Thermopneumatic Dispensing Micropump" Jounal of Sensors and Actuators.2004.05.015 , Jul-2004
Papers from Meeting Proceedings	Towe, B.C., Martineau, R., Cooney, C., Piccini, M., Daydif, P., Stout, V.,Pellis, N. "A Microfluidic Bioreporter System for Space Flight Monitoring" Proc. 25th Int Conference of IEEE Engineering in Medicine and Biology Society , Cancun, Mexico. 2003 Sep-2003
Presentation	Towe, B., Martineau, R., Cooney. C., Stout, S. Pellis, N. "A Microculture System for Space Bioreporters" NASA Annual Investigators Working Group Conference Palo Alto, California, 2004 Feb-2004